AMERICAN ACADEMY OF OPHTHALMOLOGY®

Deep Learning–Driven Exophthalmometry through Facial Photographs in Thyroid Eye Disease

Joonhyeon Park, PhD,[1,*] *Jin Sook Yoon, MD, PhD,*[2,*] *Namju Kim, MD, PhD,*[3,*] *Kyubo Shin, PhD,*[1,*] *Hyun Young Park, MD, PhD,*[2] *Jongchan Kim, MS,*[1] *Jaemin Park, MS,*[1] *Jae Hoon Moon, MD, PhD,*[1,4,5] *JaeSang Ko, MD, PhD*[1,2]

Objective: To develop and evaluate a deep learning (DL)-assisted system for proptosis measurement using facial photographs in thyroid eye disease (TED).

Design: A retrospective cohort study.

Participants: This study included 1108 patients with TED from Severance Hospital (SH) and 171 from Seoul National University Bundang Hospital (SNUBH).

Methods: The DL-assisted system was trained using 1610 facial images paired with Hertel exophthalmometry measurements from SH and externally validated using 511 SNUBH images. The system employs a dual-stream ResNet-18 neural network, utilizing both red-green-blue images and depth maps generated by the ZoeDepth algorithm.

Main Outcome Measures: Accuracy was assessed using mean absolute error (MAE), Pearson correlation coefficient, intraclass correlation coefficient (ICC), and area under the curve of the receiver operating characteristic curve.

Results: The DL-assisted system achieved an MAE of 1.27 mm for the SH dataset and 1.24 mm for the SNUBH dataset. Pearson correlation coefficients were 0.82 and 0.77, respectively, with ICCs indicating strong reliability (0.80 for SH and 0.73 for SNUBH). The receiver operating characteristic curve analysis showed area under the curves of 0.91 for SH and 0.88 for SNUBH in detecting proptosis. The system detected significant proptosis changes ($\geq$ 2 mm) with 74.6% accuracy.

Conclusions: The DL-assisted system offers an accurate, accessible method for exophthalmometry in patients with TED using facial photographs. This tool presents a promising alternative to traditional exophthalmometry, potentially improving access to reliable proptosis measurement in both clinical and nonspecialist settings.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100791 © 2025 by the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

Supplemental material available at www.ophthalmologyscience.org.

Thyroid eye disease (TED), also known as Graves' orbitopathy, is an autoimmune inflammatory disorder of the orbit associated with thyroid conditions like Graves' disease and autoimmune thyroiditis. This condition primarily affects orbital connective tissues, causing inflammation and tissue remodeling driven by activated orbital fibroblasts. Stimulated by autoantibodies and inflammatory cytokines, these fibroblasts differentiate into adipocytes and proliferate, resulting in symptoms such as periorbital swelling, eyelid retraction, diplopia, and proptosis.

Proptosis, or exophthalmos, is the forward displacement of the eyeball, commonly caused by increased volumes of extraocular muscles and fatty connective tissue within the confined bony orbit in TED. It significantly impacts quality of life, causing physical discomfort, including eye pain, dryness, redness, and irritation, making daily activities like reading and driving challenging. Moreover, the cosmetic effects of proptosis can contribute to self-consciousness, anxiety, and depression. Severe proptosis is associated with lower appearance-related quality of life.

Exophthalmometry, or proptosis measurement, involves methods such as computed tomography imaging, digital photography with image analysis, and direct measurement tools like the Hertel exophthalmometer.[1,2] The Hertel exophthalmometer is widely used for its practicality, providing a direct measurement of proptosis by measuring the distance between the lateral orbital rim and the anterior corneal surface.[3] Despite requiring specialized equipment and skill, which may affect measurement accuracy, it remains the gold standard due to its consistency and precision when performed by trained professionals.[4,5]

Assessing TED necessitates the independent evaluation of inflammatory activity and disease severity to guide appropriate treatment.[6,7] Disease activity reflects the current degree of orbital inflammation and can predict disease progression, whereas disease severity reflects the impact on patients'

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https://doi.org/10.1016/j.xops.2025.100791
ISSN 2666-9145/25

visual function and quality of life. In previous studies, we developed a machine learning–assisted system to estimate disease activity by predicting the clinical activity score (CAS) using facial photographs from digital single-lens reflex and smartphone cameras.[8,9] However, no validated tool currently exists to assess TED severity using facial photographs.

In this study, we developed a deep learning (DL)-assisted system to estimate Hertel exophthalmometry values using 1610 facial images from patients with TED. We evaluated the accuracy of these estimates and conducted external validation using an additional 511 facial images to explore the system's potential as a valuable tool for assessing TED severity.

Methods

Study Design and Dataset

This multicenter retrospective study used digital facial image datasets and medical records from patients with TED at Severance Hospital (SH) and Seoul National University Bundang Hospital (SNUBH). Patients with TED aged ≥18 years were eligible for inclusion. The SH data included 1108 patients who visited between August 2013 and July 2023. Data from 511 images of 171 patients collected at SNUBH between September 2021 and May 2024 were used for external validation. Digital facial images were captured at a 1-m distance using a digital single-lens reflex camera. Participants were instructed to keep their heads immobilized with eyes oriented in the primary position, and the head position was checked to ensure no tilting, chin-up, or chin-down postures.

This study was approved by the institutional review boards (IRB) and ethics committees of SH (IRB No. 4-2023-0518) and SNUBH (IRB No. B-2407-910-102). Informed consent was waived by the IRB due to the retrospective nature of the study. Written informed consent was obtained from 2 patients whose facial images were presented as representative cases. The study was conducted following the Declaration of Helsinki and the relevant guidelines and regulations. All images and clinical data were anonymized before being transferred outside the hospital, and the data transfer process was approved by the SH Data Review Committee (DRB No. 24-07-004). Data analysis for SNUBH was conducted within the institution.

Exophthalmometry

Proptosis was measured using a standardized Hertel exophthalmometer (Oculus Inc.) by trained oculoplastic specialists (J.S.Y. and J. Ko at SH and N.K. at SNUBH). Measurements were taken with patients seated upright, head in primary position, and examiners positioned at the same eye level in a well-lit room. The distance between the temporal orbital rims, the deepest palpable point of the angle, and the corneal apex was recorded in millimeters.[2]

Development of the DL System

Facial photographs were paired with corresponding Hertel exophthalmometer results to create a comprehensive dataset. Given the limitations of 2-dimensional photography in capturing 3-dimensional spatial information, we employed depth mapping using ZoeDepth,[10] a DL algorithm that provides depth information by assigning pixel values based on distance. Depth maps were represented in grayscale, with lighter shades indicating proximity to the camera and darker shades indicating greater distances (Fig 1). This additional dimension allowed the model to accurately interpret eye protrusion in relation to other facial features.

The model employed a dual-stream ResNet-18 neural network,[11] which processes depth maps and red-green-blue images to estimate exophthalmos by interpreting pixel data from facial images. The neural network was trained with a dataset containing facial images, depth maps, and Hertel exophthalmometer measurements, optimizing parameters to minimize the error between estimated and clinically measured degrees of exophthalmos. Additionally, exophthalmos measurements for the right eye were mirrored to create data for the left eye, effectively doubling the dataset and improving the model's generalizability across patients.

Outcome Measurements and Statistical Analyses

The main objectives were to measure exophthalmos accuracy and evaluate the impact of depth maps on prediction accuracy. The SH dataset was used with crossvalidation after being randomly split into training and test sets in a 4:1 ratio in a subject-independent manner, ensuring no individual patient appeared in both sets. The SNUBH dataset served as an external validation set. The accuracy of exophthalmos measurement was determined using the mean absolute error (MAE) for both eyes. The correlation between model predictions and actual clinical measurements was evaluated using Pearson correlation coefficient and the intraclass correlation coefficient (ICC). To assess the model's diagnostic utility for TED, we defined proptosis as an exophthalmometry value exceeding 18.3 mm, the established upper limit for the Korean population,[12–14] and evaluated the area under the curve of the receiver operating characteristic curve.

To further assess the model's accuracy and reliability, we performed a subgroup analysis based on disease severity by comparing the MAE across different degrees of proptosis and according to the presence of lid retraction. Patients were stratified into 3 groups based on Hertel exophthalmometry values: <16 mm, 16 to 20 mm, and ≥20 mm. Lid retraction was defined as a marginal reflex distance 1 of ≥6 mm, and the MAE was compared between patients with and without lid retraction. Additionally, the model's intraobserver variability was evaluated using the SNUBH external validation dataset, where 3 facial images were captured per patient during the same visit. Intraobserver variability was quantified as 1.96 × standard deviation of the difference within examiner measurements,[15] allowing us to determine the model's consistency in estimating exophthalmometry values across multiple images of the same patient under identical conditions.

We also evaluated the system's ability to detect significant exophthalmos changes, specifically those exceeding 2 mm.[16] Detection accuracy, sensitivity, and specificity were calculated to quantify the model's performance in identifying ≥2 mm changes, with detection accuracy representing the model's capacity to identify both true positives (actual changes ≥2 mm) and true negatives (changes <2 mm or no change) accurately. An ablation study further analyzed the depth maps' impact on model accuracy by comparing MAE and correlation strength in scenarios with and without depth maps. This analysis provided quantitative evidence of the depth maps' contribution to improving prediction accuracy.

All analyses were performed using R software, version 4.3.2, with a significance level of $P < 0.05$.

Results

Participant Characteristics

This study included 1108 patients from SH and 171 from SNUBH (Table S1, available at www.ophthalmologyscience.org). In the SH dataset, 758 patients visited once,

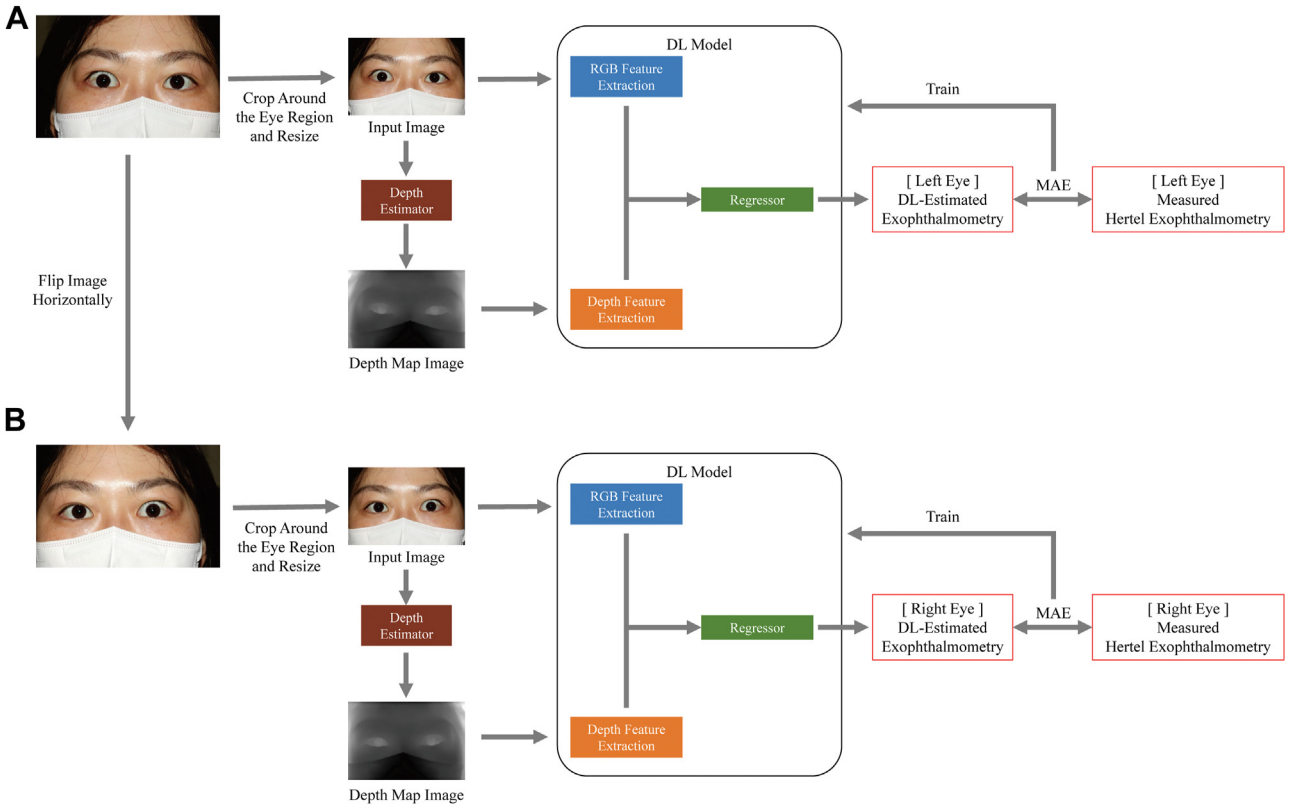


Figure 1. Workflow for DL-driven exophthalmometry using facial photographs. The model is trained to estimate the exophthalmometry value of the left eye from facial photographs. For the right eye, training is conducted with horizontally flipped images. **A,** Illustrating the training process for the left eye and **B,** for the right eye. The process begins with cropping and resizing the region around the eyes, followed by generating a depth map using the *depth estimator*, which provides 3-dimensional structural information, such as eye protrusion and surrounding geometry. Red-green-blue (RGB) images are processed by the *RGB feature extractor* to capture textural, spatial, and color-based features, whereas the *depth feature extractor* processes the depth maps to extract geometric and depth-specific characteristics. The extracted RGB and depth features are then combined and input into the *regressor*, which estimates the exophthalmometry value. The model is trained by comparing these predictions to the measured Hertel exophthalmometry values using MAE as the loss function. DL = deep learning; MAE = mean absolute error.

210 visited twice, 131 visited 3 times, 6 visited 4 times, and 3 visited 5 times, resulting in 1610 facial images with corresponding exophthalmometer measurements. Of these 1610 visits, 1295 (80.4%) were female, with an average age of 39.8 ± 11.7 years, and 1532 (95.2%) involved patients with Graves' disease. The SNUBH external validation dataset included 511 images from 171 patients; none had multiple visits. Of these, 169 patients had 3 facial images captured during a single visit, while 2 had 2 images captured. Among SNUBH patients, 129 (75.4%) were female, with an average age of 47.6 ± 16.0 years, and 98.2% were diagnosed with Graves' disease.

Accuracy of the DL System for Measuring Exophthalmos

The MAE between exophthalmometer measurements and the DL model's estimates was 1.27 mm in the SH dataset and 1.24 mm in the SNUBH dataset. Model performance was evaluated within various error margins. Within the SH dataset, 49.0% of predictions were within 1 mm, 65.8% within 1.5 mm, and 79.3% within 2 mm of actual measurements. For the SNUBH dataset, 48.8% of predictions were within 1 mm, 67.2% within 1.5 mm, and 77.5% within 2 mm of actual measurements (Fig 2). The Pearson correlation coefficient was 0.82 ($P < 0.01$; 95% confidence interval [CI], 0.81−0.83) for SH and 0.77 ($P < 0.01$; 95% CI, 0.73−0.81) for SNUBH (Fig 3). The ICC indicated excellent reliability, with values of 0.80 ($P < 0.001$; 95% CI, 0.79−0.81) for SH and 0.73 ($P < 0.001$; 95% CI, 0.68−0.78) for SNUBH. Diagnostic performance for detecting proptosis, assessed via receiver operating characteristic curves, yielded an area under the curve of 0.91 (95% CI, 0.90−0.92) for SH and 0.88 (95% CI, 0.86−0.93) for SNUBH (Fig 4). Table 2 summarizes the sensitivity, specificity, accuracy, positive predictive value, and negative predictive value of exophthalmos detection across both datasets.

In the subgroup analysis by severity of proptosis, the MAE varied across severity groups (Table S3, available at www.ophthalmologyscience.org). The model demonstrated the highest accuracy in the 16 to 20 mm range while showing a tendency to overestimate in cases with mild

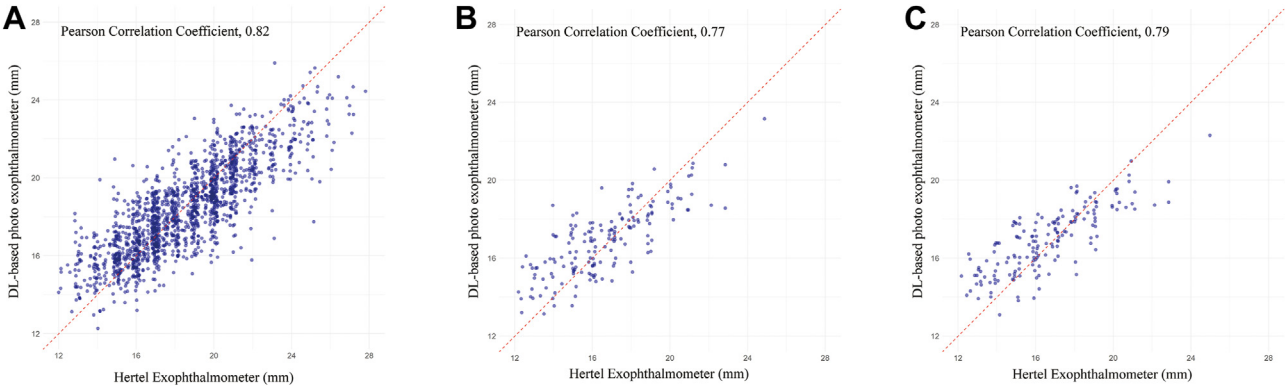


Figure 2. Correlation between proptosis values predicted by a DL model and those measured using a Hertel exophthalmometer. **A,** Severance Hospital dataset: 1610 images of 1108 patients analyzed. **B,** Seoul National University Bundang Hospital dataset (1 image): 171 images of 171 patients analyzed. **C,** Seoul National University Bundang Hospital dataset (3 images): 507 images of 169 patients with 3 facial images analyzed. The mean proptosis measurement from the 3 images was compared with exophthalmometer values. DL = deep learning.

proptosis and underestimate in cases with severe proptosis, suggesting a regression-to-the-mean effect. In the subgroup analysis by presence of lid retraction, the MAE in the SH and SNUBH datasets was 1.34 mm and 1.09 mm in patients with lid retraction, compared with 1.26 mm and 1.15 mm in patients without lid retraction, respectively.

For SH patients with multiple visits, the model's capacity to detect proptosis changes exceeding 2 mm showed an accuracy of 74.6%, with a sensitivity of 70.0% and specificity of 82.3%.

Among 169 SNUBH patients with 3 facial images captured in 1 visit, the mean estimated exophthalmometry value obtained from the 3 images was compared with exophthalmometer measurements, showing an MAE of 1.23 mm, a Pearson correlation coefficient of 0.79 ($P < 0.01$; 95% CI, 0.76–0.83), and an ICC of 0.73 ($P < 0.01$; 95% CI, 0.67–0.78). The receiver operating characteristic curve for exophthalmos detection showed an area under the curve of 0.89 (95% CI, 0.86–0.93). In this analysis, 50.0% of estimates were within 1 mm, 66.6% within 1.5 mm, and 79.0% within 2 mm of actual measurements (Fig 2). Furthermore, the intraobserver variability of the model, assessed using 3 facial images captured in a single visit, was determined to be 0 ± 0.42 mm.

Ablation Study

The ablation study evaluated the model's accuracy with and without ZoeDepth-generated depth maps. For the SH dataset, the MAE without depth maps was 1.35 mm, and the Pearson correlation coefficient was 0.79 ($P < 0.01$; 95% CI, 0.78–0.80). With depth maps, the MAE improved to 1.27 mm, and the Pearson correlation coefficient increased to 0.82 ($P < 0.01$; 95% CI, 0.81–0.83).

For the SNUBH dataset, analyzing a single image per patient without depth maps yielded an MAE of 1.45 mm and a Pearson correlation coefficient of 0.71 ($P < 0.01$; 95% CI, 0.65–0.76). Incorporating depth maps improved the MAE to 1.24 mm and correlation to 0.77 ($P < 0.001$; 95% CI, 0.73–0.81).

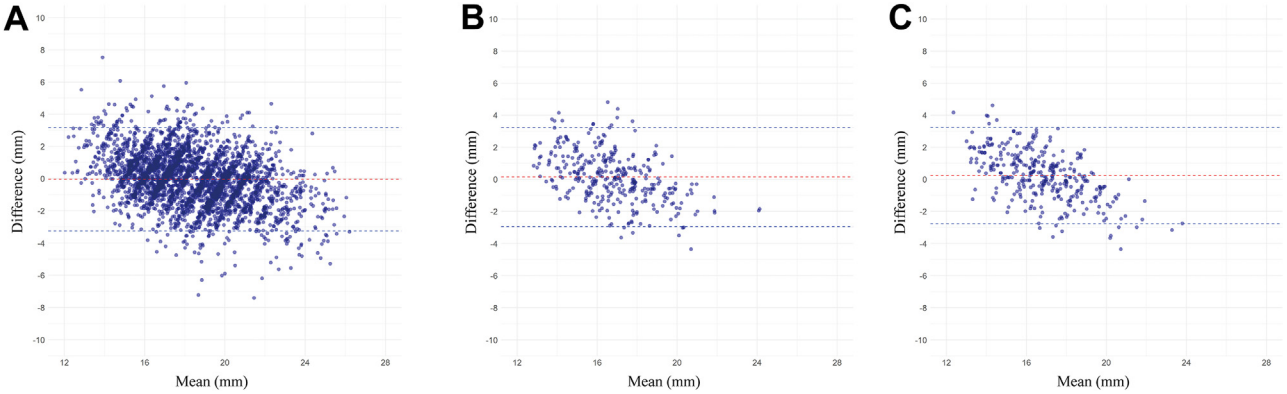


Figure 3. Bland–Altman analysis comparing protrusion measurements: Hertel exophthalmometer vs. deep learning model predictions. **A,** Severance Hospital dataset: 1610 images of 1108 patients analyzed. **B,** Seoul National University Bundang Hospital dataset (1 image): 171 images of 171 patients analyzed. **C,** Seoul National University Bundang Hospital dataset (3 images): 507 images of 169 patients with 3 facial images analyzed. The mean proptosis measurement from the 3 images was compared with exophthalmometer values.

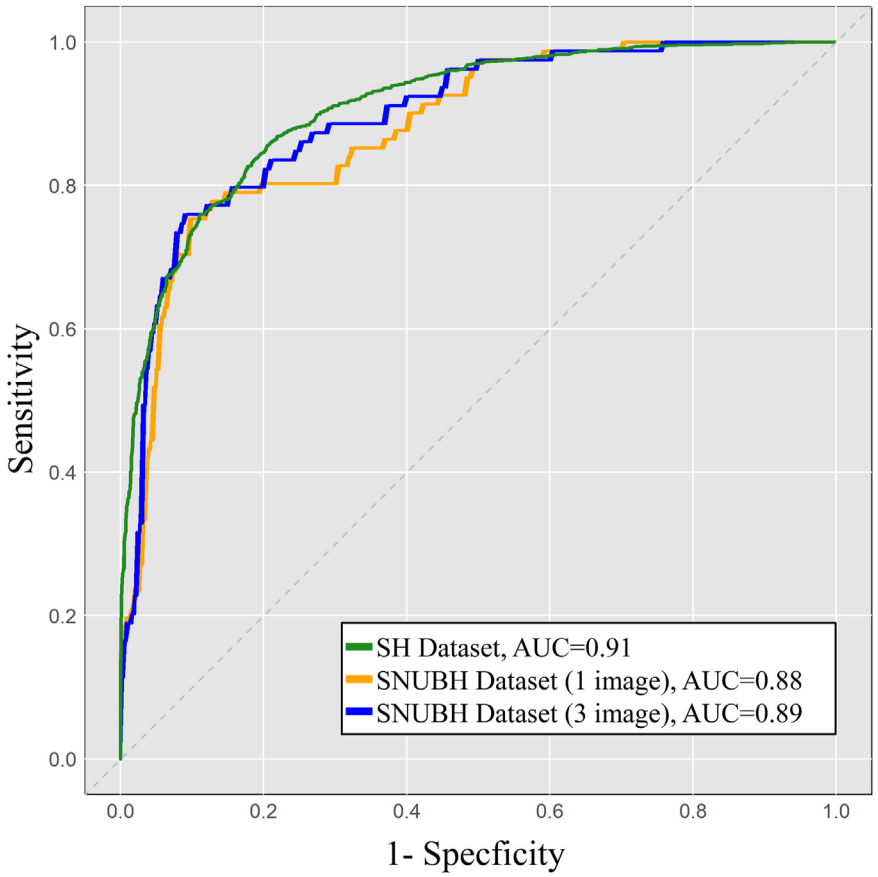


Figure 4. Receiver operating characteristic curve for proptosis detection derived from the deep learning system. Proptosis is defined as an exophthalmometry value exceeding 18.3 mm, the upper limit of the normal range for the Korean population. Severance Hospital dataset (green line): 1610 images of 1108 patients analyzed. Seoul National University Bundang Hospital dataset (1 image) (yellow line): 171 images of 171 patients analyzed. Seoul National University Bundang Hospital dataset (3 images) (blue line): 507 images of 169 patients with 3 facial images analyzed. The mean proptosis measurement from the 3 images was used in the analysis. AUC = area under the curve; SH = Severance Hospital; SNUBH = Seoul National University Bundang Hospital.

When analyzing 3 images per patient in the SNUBH dataset, averaging values without depth maps resulted in an MAE of 1.53 mm and a Pearson correlation coefficient of 0.71 ($P < 0.01$; 95% CI, 0.65−0.76). With depth maps, the MAE improved to 1.23 mm and the Pearson correlation coefficient to 0.79 ($P < 0.01$; 95% CI, 0.76−0.83).

Discussion

This study introduced a DL-based system for estimating proptosis in patients with TED using facial photographs. The model was trained on 1610 facial images paired with corresponding Hertel exophthalmometry values and

Table 2. Performance Metrics of Proptosis Detection

Dataset	Accuracy (%, 95% CI)	Sensitivity (%, 95% CI)	Specificity (%, 95% CI)	PPV (%, 95% CI)	NPV (%, 95% CI)
SH dataset*	82.4 (81.0−83.7)	82.7 (80.9−84.5)	82.2 (80.3−84.0)	81.2 (79.1−83.2)	83.6 (81.7−85.4)
SNUBH dataset (1 image)†	86.6 (82.8−89.8)	65.4 (55.1−75.3)	93.1 (90.0−96.1)	74.7 (63.9−84.8)	89.7 (86.0−93.0)
SNUBH dataset (3 images)‡	87.9 (84.3−91.4)	67.1 (56.4−77.8)	94.2 (91.3−96.9)	77.9 (67.8−87.1)	90.4 (86.5−93.9)

CI = confidence interval; NPV = negative predictive value; PPV = positive predictive value; SH = Severance Hospital; SNUBH = Seoul National University Bundang Hospital.
*One thousand six hundred ten images of 1108 patients were analyzed in the SH dataset.
†One hundred seventy-one images of 171 patients were analyzed in the SNUBH dataset (1 image).
‡Five hundred seven images of 169 patients with 3 facial images captured in a single visit were analyzed in the SNUBH dataset (3 images). The mean estimated exophthalmometry value from the 3 images was used to detect proptosis.

externally validated with data from 171 independently sourced patients. Although widely used, Hertel exophthalmometers exhibit notable interevaluator variability, with agreement within 1 mm between clinicians' measurements ranging from 30% to 80%.[17–19] In comparison, our DL-based model estimated values within 1 mm of Hertel measurements approximately 50% of the time. At a 2 mm threshold, concordance rates for Hertel measurements range from 74% to 82%,[18,20] closely aligning with the 78% to 79% achieved by our model. Regarding ICC, Hertel measurements have reported ICCs ranging from 0.57 to 0.90[1,21]; our model demonstrated an ICC of 0.73 to 0.80, indicating reliability comparable to the interobserver variability of Hertel exophthalmometry. Additionally, the intraobserver variability of our model, assessed using 3 facial images captured in a single visit, was 0 ± 0.42 mm, which is lower than the reported Hertel intraobserver variability (0.56–0.61 mm), suggesting greater measurement consistency.[15] External validation showed that averaging estimates from 3 photographs taken at the same time improved accuracy, suggesting that utilizing multiple images could further enhance this DL system's precision in clinical practice.

For TED management, evaluating disease activity and severity is critical as these guide treatment decisions, which may involve immunomodulatory agents or surgery, both come with significant side effects.[6,22,23] Severity assessment is more challenging than disease activity and requires a multifactorial approach, including proptosis, ocular movement, lid retraction, and soft-tissue involvement. Accurate proptosis measurement often requires specialized training and equipment, complicating treatment decisions and response tracking. The CAS is widely used for TED activity assessment, but its accuracy may vary depending on the evaluator's expertise.[8,24] To address these challenges, we developed and validated a DL system using facial images to predict CAS.[8,9] However, CAS alone may be unreliable in certain clinical scenarios, particularly in patients with low CAS scores who may still experience disease progression or a poor prognosis (Fig S5, available at www.ophthalmologyscience.org).[7,25] This limitation highlights the need for a more comprehensive approach to evaluating TED activity that integrates objective markers, such as proptosis, with disease progression history to ensure accurate assessment and informed treatment decisions.[6]

Proptosis measurement is crucial not only for assessing TED severity but also for diagnosing and evaluating other ophthalmic conditions, such as orbital tumors, trauma, and inflammation. Although Hertel exophthalmometry is widely used, it can be inaccurate in certain situations, showing significant interobserver and intraobserver variability,[1–3,15,18–21] particularly with less experienced practitioners.[2,15,19,21] Although primary care physicians and endocrinologists play a vital role in the initial diagnosis and referral for TED,[6] expecting them to obtain precise Hertel measurements is often unrealistic. Computed tomography offers relatively accurate proptosis measurements significantly correlated with Hertel values,[1,17,26,27] but its high cost, scheduling difficulties, and radiation exposure limit its routine use in TED diagnosis and monitoring.

Previous studies explored digital imaging techniques for proptosis measurement. One method involves capturing sagittal facial photographs with a reference object of known length to measure the distance from the lateral orbital rim to the corneal apex.[28,29] However, this technique requires assistance in capturing images and calculations that may be inaccurate without detailed knowledge of facial anatomy. Another approach, limited to specific smartphone models with specialized infrared cameras, employs geometric facial analysis for exophthalmometry.[30] This method, although innovative, requires precise marking of the lateral orbital rim and complex analysis, posing challenges for patients and primary care physicians without specialized training. Our study's results, achieving an ICC of 0.80, exceeded the ICC of 0.63 reported by this method.[30] Although promising, these methods face practical and technical challenges that limit their clinical utility.

A major advantage of this study is its simplicity and accessibility, eliminating the need for specialized equipment or extensive training. Our system enables nonexperts to upload frontal facial photographs, allowing the DL algorithm to analyze these images and estimate proptosis values reasonably comparable to Hertel measurements. Notably, the model maintained stable performance even in patients with significant lid retraction, which can otherwise give the false impression of proptosis by increasing ocular surface exposure during visual assessments (Fig S5). Patients can also use selfies for serial measurements, facilitating longitudinal monitoring of proptosis progression. Moreover, this method enables retrospective proptosis estimations from previously captured photographs, allowing clinicians to quantify proptosis changes due to TED and establish appropriate treatment targets. Consequently, this approach broadens access to accurate proptosis measurements, enhancing the ability to monitor disease progression and therapeutic responses.

The study's main limitation is potential bias from using the Hertel exophthalmometer as a reference to train the DL model. Given the known variability in Hertel measurements, these errors may have been propagated into the DL process, potentially limiting accuracy improvements over traditional Hertel measurements. To mitigate this, we utilized measurements taken exclusively by highly experienced oculoplastic faculty and conducted external validation. Additionally, the model showed minor estimation biases in extreme cases, overestimating mild proptosis and underestimating severe proptosis. Another significant limitation is that this study was conducted exclusively in the Korean population. Given the nature of facial image analysis, there may be racial differences in performance. Therefore, follow-up studies across diverse ethnic groups are underway to enhance generalizability. Additionally, as this study used digital single-lens reflex images captured in a controlled setting, further validation is necessary to confirm that smartphone-captured images yield reliable results for self-monitoring by patients, particularly given differences in device color calibration algorithms.

In conclusion, this study introduced a DL-assisted system for proptosis measurement in patients with TED using digital facial images. This system offers a promising alternative that is accessible and easy to use, even for

nonexperts. Despite potential biases and the need for further validation across diverse populations, this method democratizes access to accurate proptosis measurements, facilitating improved TED severity assessment, disease progression monitoring, and treatment response evaluation. Future studies should aim to address these limitations and refine the system to accurately assess proptosis in other conditions, broadening its applicability beyond TED.

Data Availability Statement

Due to ethical and legal restrictions imposed by the IRBs of the participating institutions, raw data, including medical records and facial images, are not available upon request. Deidentified data are available to researchers who meet the criteria for accessing confidential data and enter into a data usage agreement with the institutions.

Footnotes and Disclosures

Originally received: December 6, 2024.
Final revision: February 21, 2025.
Accepted: March 31, 2025.
Available online: April 8, 2025. Manuscript no. XOPS-D-24-00545.

[1] Division of Research & Development, THYROSCOPE Inc., Ulsan, Republic of Korea.

[2] Department of Ophthalmology, Severance Hospital, Institute of Vision Research, Yonsei University College of Medicine, Seoul, Republic of Korea.

[3] Department of Ophthalmology, Seoul National University Bundang Hospital and Seoul National University College of Medicine, Seongnam, Republic of Korea.

[4] Department of Internal Medicine, Seoul National University Bundang Hospital and Seoul National University College of Medicine, Seongnam, Republic of Korea.

[5] Center for Artificial Intelligence in Healthcare, Seoul National University Bundang Hospital, Seongnam, Republic of Korea.

*J.P, J.S.Y, N.K and K.S contributed equally to this study.

Disclosure(s):

All authors have completed and submitted the ICMJE disclosures form.

The author(s) have made the following disclosure(s):

Jaemin Park: Financial support – THYROSCOPE INC. (CEO), Ministry of SMEs and Startups (Republic of Korea) (research grant), Ministry of Science, ICT (Republic of Korea) (research grant); Patents – THYROSCOPE INC. (the company has a pending patent application for the DL-assisted system to estimate eye protrusion); Leadership – THYROSCOPE INC. (CEO); Stocks – THYROSCOPE INC.

J. Ko: Financial support – THYROSCOPE INC. (research director); Stocks – THYROSCOPE INC. (holds stock options).

J.H.M.: Financial support – THYROSCOPE INC. (board member); Leadership – THYROSCOPE INC. (board member); Stocks – THYROSCOPE INC.

J. Kim: Financial support – THYROSCOPE INC. (employee), Ministry of Science, ICT (Republic of Korea) (research grant); Patents – THYROSCOPE INC. (the company has a pending patent application for the DL-assisted system to estimate eye protrusion); Stocks– THYROSCOPE INC. (holds stock options).

J.S.Y.: Financial support – THYROSCOPE INC. (consultant).

K.S.: Financial support – THYROSCOPE INC. (board member), Ministry of Science, ICT (Republic of Korea) (research grant); Leadership – THYROSCOPE INC. (board member); Stocks – THYROSCOPE INC. (holds stock options).

N.K.: Financial support – THYROSCOPE INC. (consultant); Stocks – THYROSCOPE INC. (holds stock options).

This work was supported by the Technological Innovation R&D Program (RS-2023-00281098) funded by the Ministry of SMEs and Startups (MSS, Korea) and a grant from the AI-based medical system digital transformation support program through the National IT Industry Promotion Agency (NIPA), funded by the Ministry of Science, ICT (No. H0906-24-1006).

Support for Open Access publication was provided by the Ministry of SMEs and Startups (MSS, Korea).

HUMAN SUBJECTS: Human subjects were included in this study. This study was approved by the institutional review boards (IRB) and ethics committees of Severance Hospital (IRB No. 4-2023-0518) and Seoul National University Bundang Hospital (IRB No. B-2407-910-102). Informed consent was waived by the IRB due to the retrospective nature of the study. Written informed consent was obtained from 2 patients whose facial images were presented as representative cases. The study was conducted following the Declaration of Helsinki and the relevant guidelines and regulations.

No animal subjects were included in this study.

Author Contributions:

Conception and design: Yoon, Namju Kim, Shin, Moon, Ko

Data collection: Joonhyeon Park, Yoon, Namju Kim, Shin, Hyun Young Park, Jongchan Kim, Moon, Ko

Analysis and interpretation: Joonhyeon Park, Yoon, Shin, Jongchan Kim, Jaemin Park, Moon, Ko

Obtained funding: Jaemin Park

Overall responsibility: Joonhyeon Park, Yoon, Namju Kim, Shin, Hyun Young Park, Jongchan Kim, Jaemin Park, Moon, Ko

Abbreviations and Acronyms:
CAS = clinical activity score; **CI** = confidence interval; **DL** = deep learning; **ICC** = intraclass correlation coefficient; **IRB** = institutional review board; **MAE** = mean absolute error; **SH** = Severance Hospital; **SNUBH** = Seoul National University Bundang Hospital; **TED** = thyroid eye disease.

Keywords:
Deep learning, Exophthalmometry, Thyroid eye disease, Proptosis, Dual-stream ResNet-18.

Correspondence:
JaeSang Ko, MD, PhD, Department of Ophthalmology, Severance Hospital, Institute of Vision Research, Yonsei University College of Medicine, 50-1, Yonsei-ro, Seodaemun-gu, Seoul 03722, Republic of Korea. E-mail: jaesangko@yonsei.ac.kr; and Jae Hoon Moon, MD, PhD, Department of Internal Medicine, Seoul National University Bundang Hospital, Seoul National University College of Medicine 82, Gumi-ro 173 Beon-gil, Bundang-gu, Seongnam-si, Gyeonggi-do 13620, Republic of Korea. E-mail: jaehoon.moon@snu.ac.kr.

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